

Optical Compute Interconnects (OCI) MSA

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On behalf of the OCI MSA

Lightning Round Questions Posed

- **What problem is the (OCI) MSA solving?**
 - Definition of an Interconnect which is
 - Power Efficient
 - Multi-vendor Interoperability
 - Multi-generational, Rapid Scaling
- **Why all the existing standards bodies and consortia are not solving that problem?**
- **What the MSA promises to actually produce?**
- **How the MSA will avoid stifling innovation?**
- **How companies can join the MSA?**

OCI-MSA



Founding Members

Industry alignment around an optical scale-up specification.



Est. March 2026

Future AI infrastructure built on open standards.



Optical Scale-Up

Architecture that delivers the efficiency required to scale AI infrastructure.

Move Scale-up from In-Rack to Multi-Rack & Multi-Row

For more information and the OCI-MSA GEN1 Specification please see www.oci-msa.org

OCI MSA Implementation

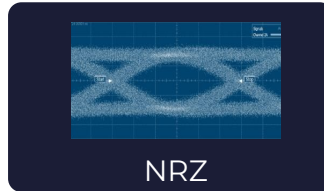
OPTICAL IMPLEMENTATION FOR SCALE-UP

Optical Scale-up only
limited by RADIX



Est. March 2026

Scalable, Lowest
Power



Standard Lasers



Reduce Cost w/o
RADIX impact



- Lower Power
- Lower Cost
- Lower Latency
- High Bandwidth per Fiber

Enable Scale-up from In-Rack to Multi-Rack & Multi-Row

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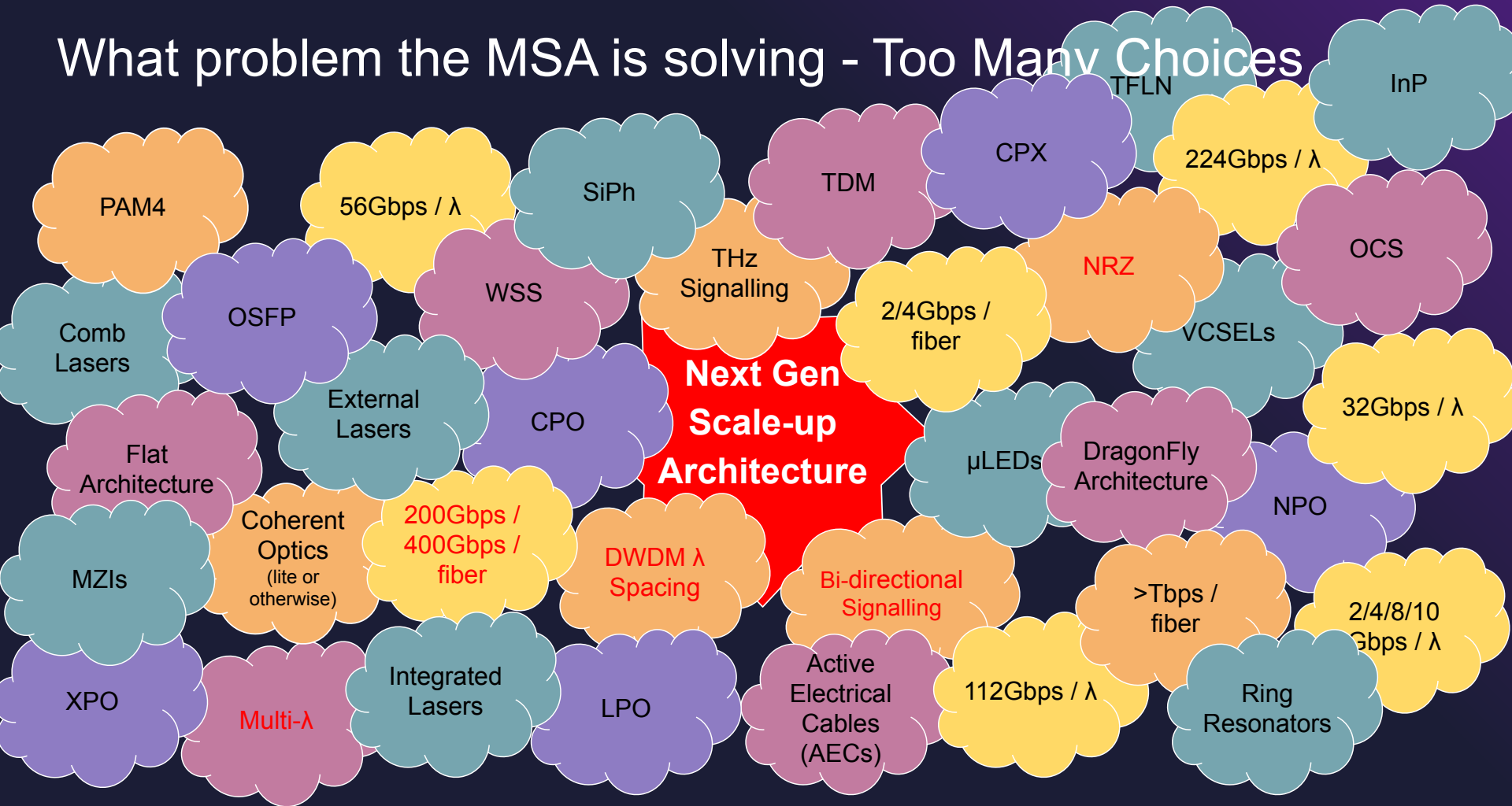
What problem the MSA is solving: Scaling

Bandwidth	Spec Timeline	Electrical	Optical	Linear	Comment
200Gbps	Today	100Gbaud PAM4	100Gbaud PAM4	YES	
400Gbps	2027 (?)	PAM4/6 PAM8 (?), SBD (?)	200Gbaud PAM4	Maybe (?)	IC area & pJ/b diminishing returns Maybe cabled backplane support
800Gbps	>2030 (?)	?	?	Unlikely	
OCI GEN1	Today	N/A, wide parallel	4λ x 50Gbps NRZ, DWDM, bidi	NA	GEN2 spec @ 400Gbps = 2027 GEN3 spec @ 800Gbps = 202x

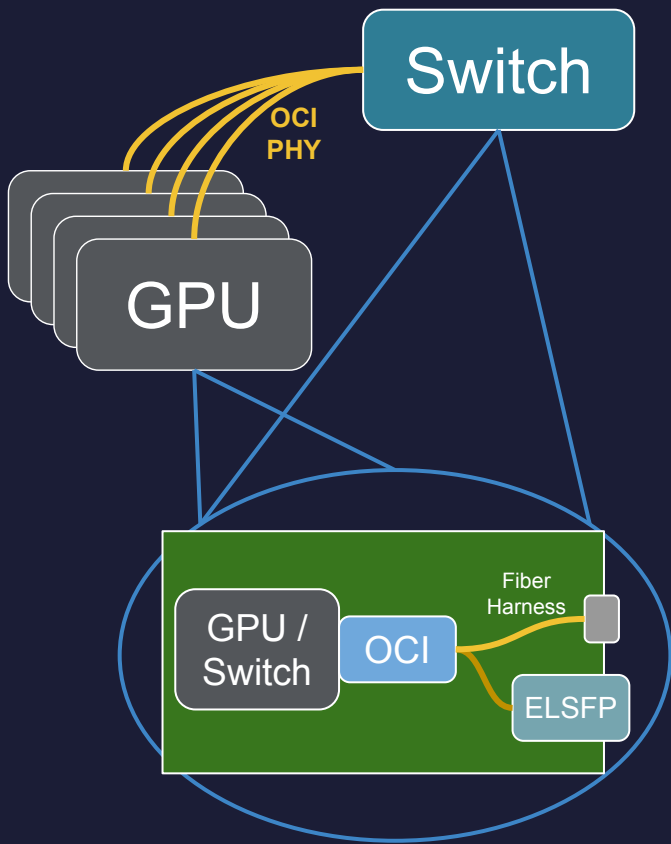
SBD = Simultaneous Bi-Directional Signaling

- AI pace of innovation likely faster than “standard” Ethernet AUI interfaces
 - 400Gbps & 800Gbps innovations driving power and efficiency in the wrong direction
 - OCI multi-generational scaling supports required pace of AI cluster innovation
- Future larger AI cluster scale-up domains will require highest radix switches
 - Larger number of lower BW links can support larger scale-up domain sizes

What problem the MSA is solving - Too Many Choices



What problem the MSA is solving: Ecosystem



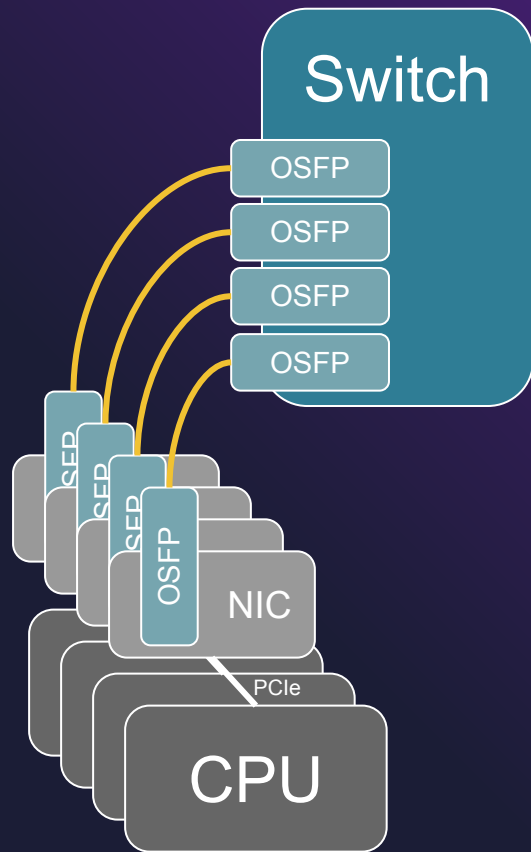
Networks require an interoperable interconnect

Interoperability for integrated optics requires:

- Common PHY spec (OCI)
- ~~Proprietary deployment~~

Additionally

- “Forced Marriages” don’t work
- Component Ecosystems are also necessary



Lightning Round Questions Posed

- **What problem is the (OCI) MSA solving?**
- **Why all the existing standards bodies and consortia are not solving that problem?**
 - Rate of innovation, power/cost efficiency, distributed focus
- **What the MSA promises to actually produce?**
 - We're producing specifications to enable interoperability between end points
 - Implementation details will be addressed elsewhere in the ecosystem
- **How the MSA will avoid stifling innovation?**
 - We're enabling technical innovation by not dictating physical implementation details
- **How companies can join the MSA?**
 - Still figuring this out ourselves, balance of speed vs inclusivity